

Chapter 3-3

Analog Transmission

Topics

- Digital Data-to -Analog Signal
- Analog Signal-to-Analog Signal
- Multiplexing

Analog transmission

- Analog –continuous signal that is having a value at every instance of time.
- Analog transmission is the transmission of analog signal over a transmission media.
- Digital transmission is very desirable but a low-pass channel with a very large bandwidth is needed.
- Analog transmission is the only choice if we have a band-pass channel

Digital-to-Analog Conversion

- By changing one characteristic(amplitude, frequency, or phase) of a simple electric signal(analog signal) based on the information in the digital signal(0s or 1s)
- Transmitting data from one computer (digital) to another across a public phone line (carrier analog signals) requires that the digital data must be modulated on an analog signal that has been manipulated to loo two distinct values corresponding to binary 0 and 1
- main use is public telephone system
 - has freq range of 300Hz to 3400Hz
 - use modem (modulator-demodulator)

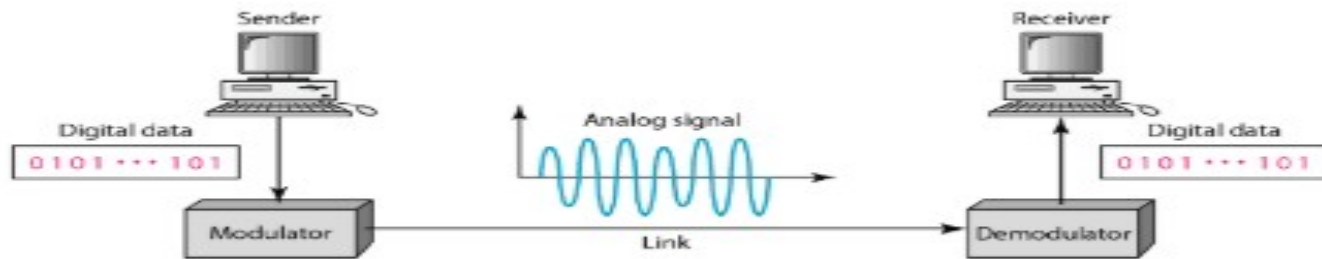
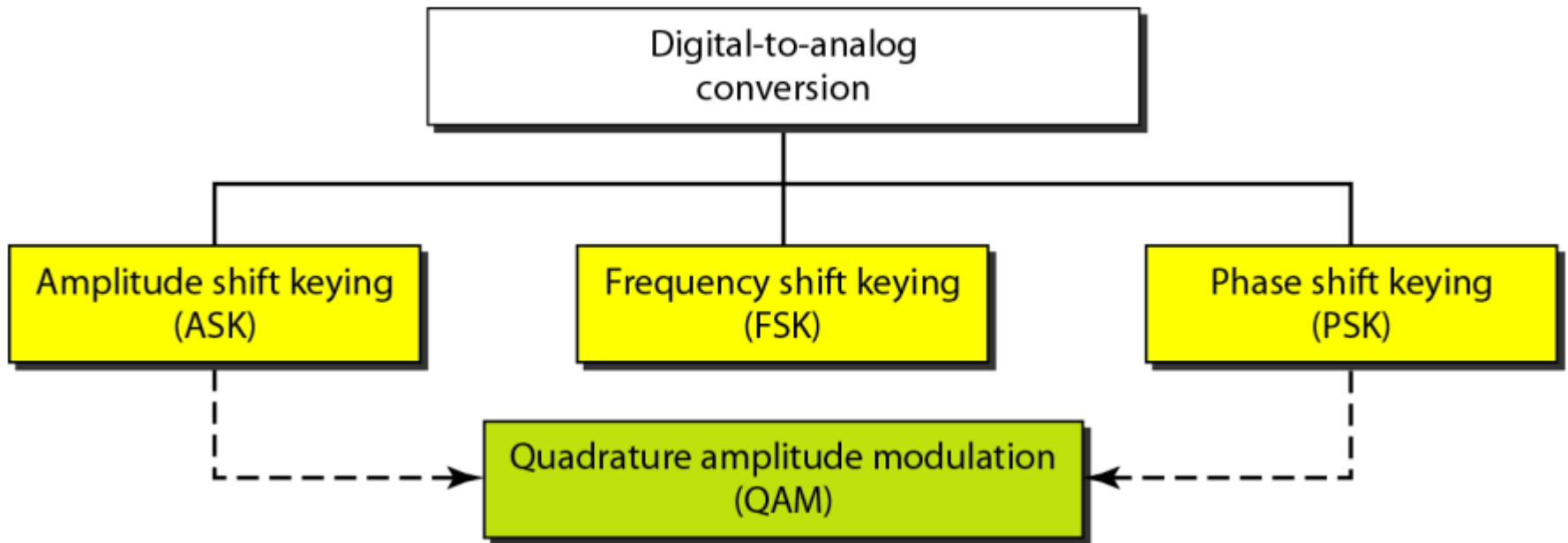


Figure shows the relationship between the digital information, the digital-to-analog modulating process, and the resultant analog signal.

Digital Data, Analog Signal



- **Aspects of Digital-to-Analog Conversion**
 - bit
 - baud rates
 - carrier signal.

- ***Data Element Versus Signal Element***

- Data element as the smallest piece of information to be exchanged, the bit.
- signal element smallest unit of a signal that is constant.

- ***Data Rate Versus Signal Rate***

- Bit rate/data rate is the number of bits per second.
- Baud rate / signal rate is the number of signal elements per second.
- In the transmission of digital data, the baud rate is less than or equal to the bit rate.

- **A signal unit is composed of one or more bits.**
- The fewer signal units required, the more efficient the system and less bandwidth required to transmit more bits.
- The baud rate determines the **bandwidth** required to send the signal NOT the number of bits.
- In transportation, a **baud** is analogous to a car, and a **bit** is analogous to a passenger. A car can carry one or more passengers.
- The number of cars not the number of passengers determines the traffic and therefore the need for wider highways.

- We can define the data rate (bit rate) and the signal rate (baud rate). The relationship between them is **$S = N/r \text{ baud}$** where N is the data rate (bps) and r is the number of data elements carried in one signal element. The value of r in analog transmission is $r = \log_2 L$, where L is the type of signal element, not the level.

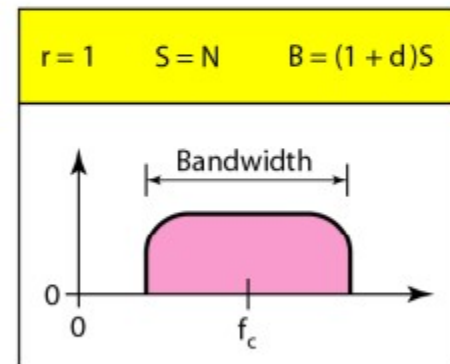
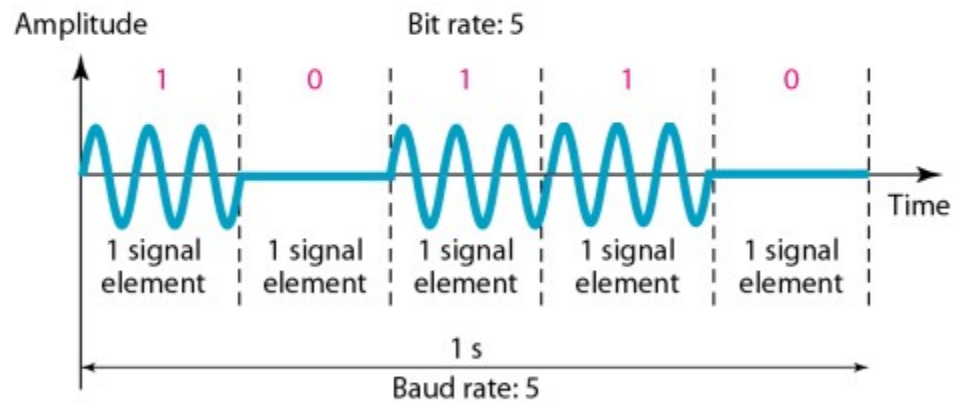
- ***Carrier Signal***

- In **analog transmission**, the sending device produces a **high-frequency signal** that acts as a base for the information signal.
- This base signal is called the **carrier signal** or **carrier frequency**.
- The receiving device is tuned to the frequency of the carrier signal that it expects from the sender.
- Digital information then changes the carrier signal by modifying one or more of its characteristics (amplitude, frequency, or phase).
- This kind of modification is called **modulation** or **shift keying** and the information signal is called the **modulating signal**.

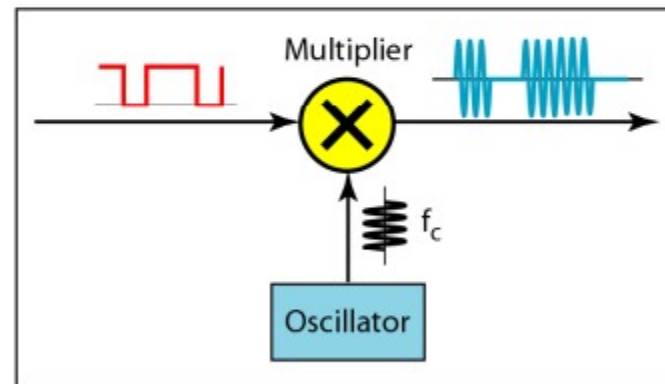
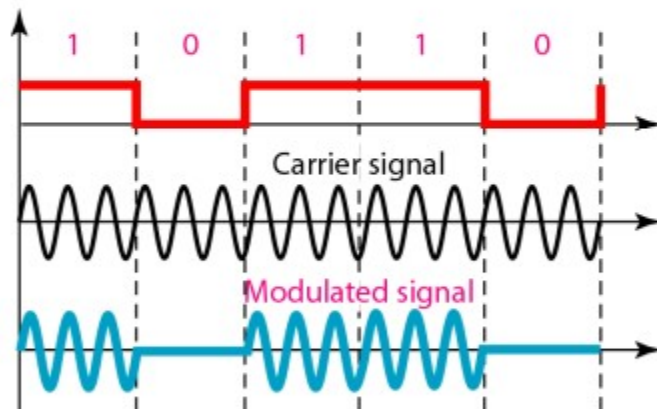
Amplitude Shift Keying

- The amplitude (height) of the carrier signal is varied to represent binary 0 or 1. Both frequency and phase remain constant while the amplitude changes.
- Although we can have several levels (kinds) of signal elements, each with a different amplitude, ASK is normally implemented using only two levels. This is referred to as **binary amplitude shift keying** or ***on-off keying (OOK)***.
- The peak amplitude of one signal level is 0; the other is the same as the amplitude of the carrier frequency.

Binary amplitude shift keying



Implementation of binary ASK



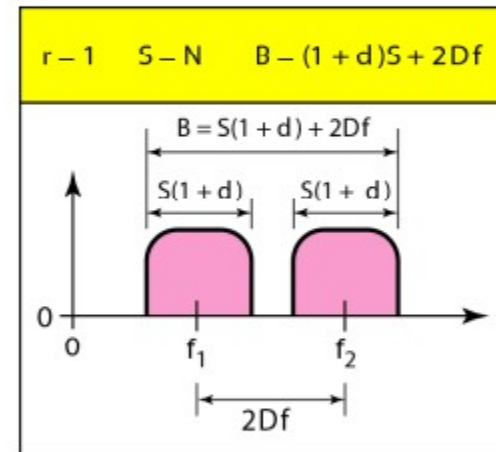
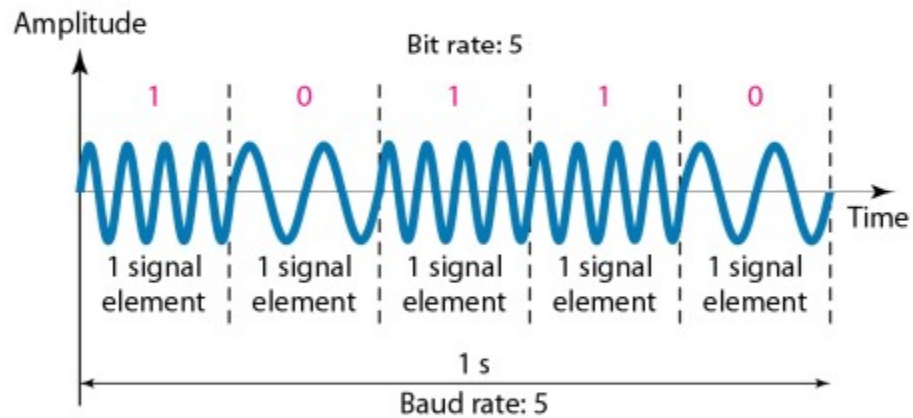
- susceptible to noise interference because noise mainly affects amplitude
- inefficient
- used for
 - up to 1200bps on voice grade lines
 - very high speeds over optical fiber

Frequency Shift Keying

- In frequency shift keying, the frequency of the carrier signal is varied to represent binary 1 or zero.
- The frequency of the modulated signal is **constant** for the duration of one signal (each bit duration) element, but changes for the next signal element if the data element changes.
- Both peak amplitude and phase remain constant for all signal elements

- most common is binary FSK (BFSK)
- two binary values represented by two different frequencies (near carrier)
- less susceptible to error than ASK
- used for
 - up to 1200bps on voice grade lines
 - high frequency radio
 - even higher frequency on LANs using co-ax

Binary frequency shift keying



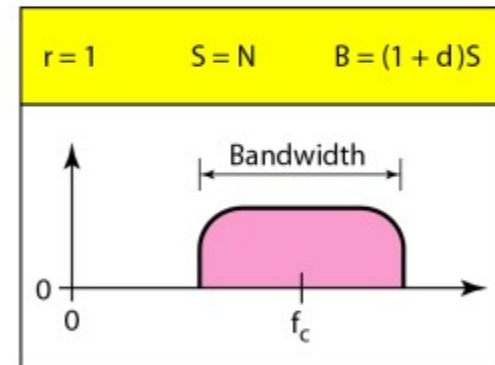
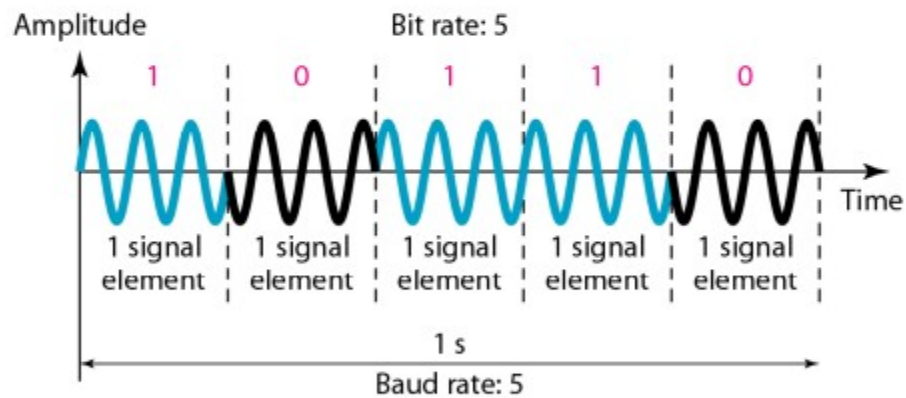
Multiple FSK

- each signalling element represents more than one bit
- more than two frequencies used
- more bandwidth efficient
- more prone to error

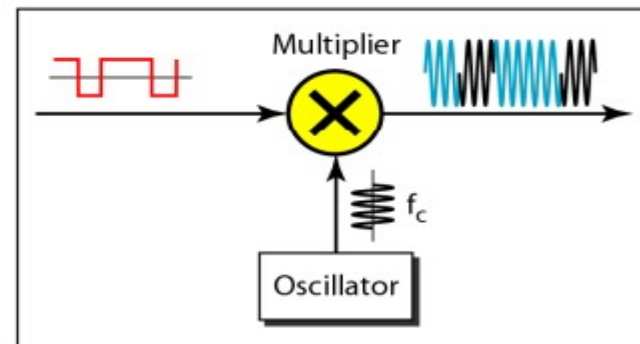
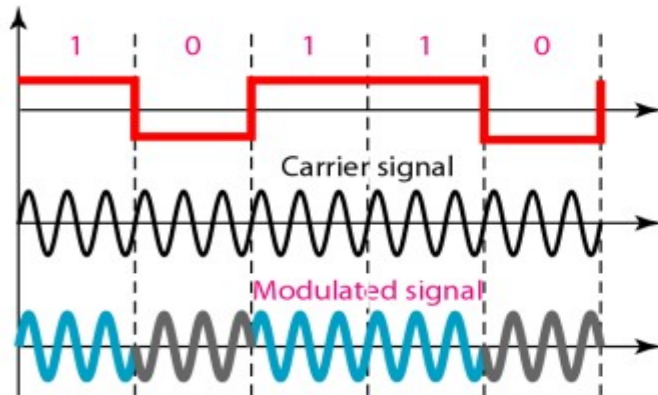
Phase Shift Keying

- The phase of the carrier is varied to represent to represent binary 1 or 0.
- The phase of the signal during each bit duration is constant, and its value depends on the bit (0 or 1).
- Both peak amplitude and frequency remain constant as the phase changes.
- Today, PSK is more common than ASK or FSK.
- The simplest PSK is **binary PSK**, in which we have only two signal elements, one with a phase of 0° , and the other with a phase of 180° to represent the two binary digits.

Binary phase shift keying



Implementation of BASK



- A constellation or phase-state diagram shows the relationship between phase and bit value.
- PSK is **not susceptible** to the **noise degradation** that affects ASK or to the **bandwidth** limitations of FSK.
- Therefore, we can use four variations of a signal, each representing 2 bits. This technique is called 4-PSK or Q-PSK. The pair of bits represented by each phase is called a **dibit**.

- The 4-PSK idea can be extended to 8-PSK. Instead of 90° , we can vary the signal by shifts of 45° .
- With eight different phases, each shift can represent 3 bits (a tribit).
- The minimum bandwidth required for PSK transmission is the same as that required for ASK transmission (for the same reasons).
- While the maximum baud rates of ASK and PSK are the same for a given bandwidth, PSK bit rates using the same bandwidth can be 2 or more times greater.

PSK constellation

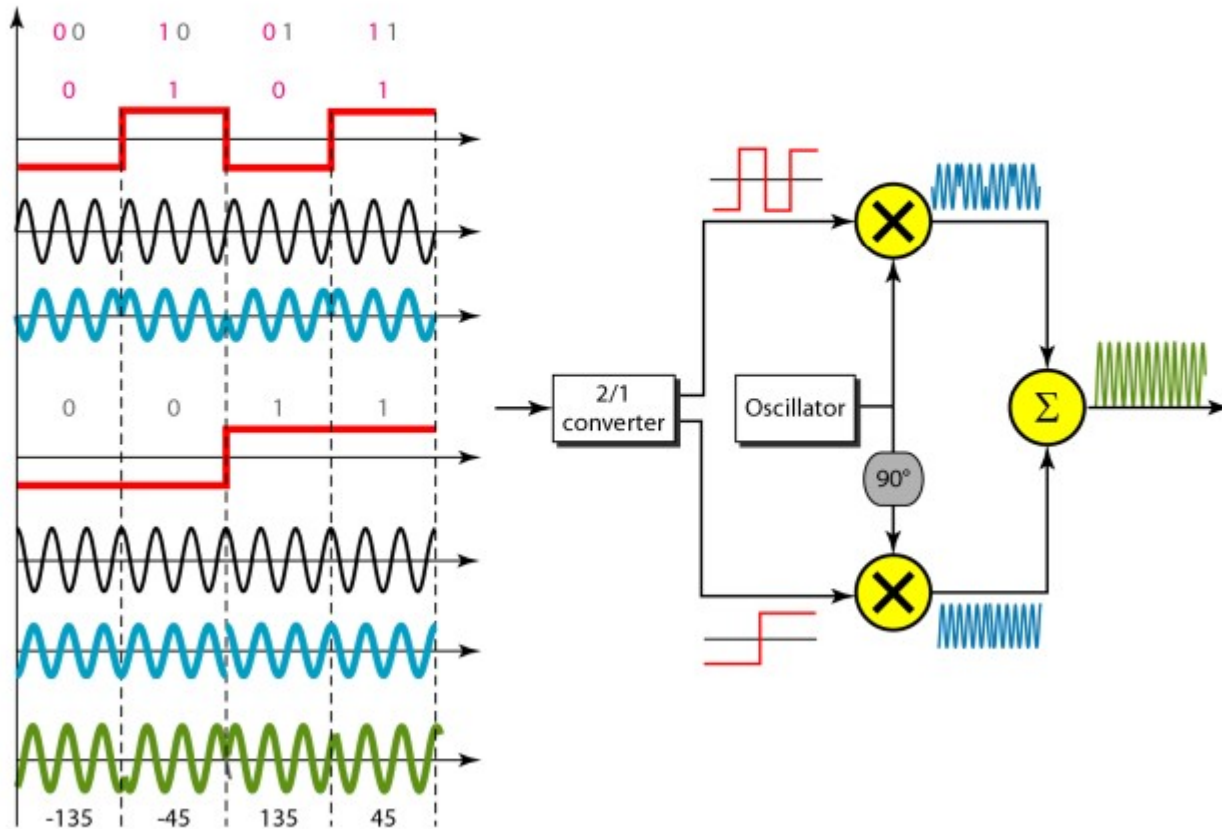
Bit	Phase
0	0
1	180

Bits



Constellation diagram

QPSK and its implementation



Quadrature Amplitude Modulation (QAM)

- PSK is limited by the ability of the equipment to distinguish small differences in phase.
- So far we have been altering only one of the three characteristics of a sine wave at a time. What if we alter two?
- Bandwidth limitations make combinations of FSK with other changes practically useless.
- As such, we will combine ASK with PSK.

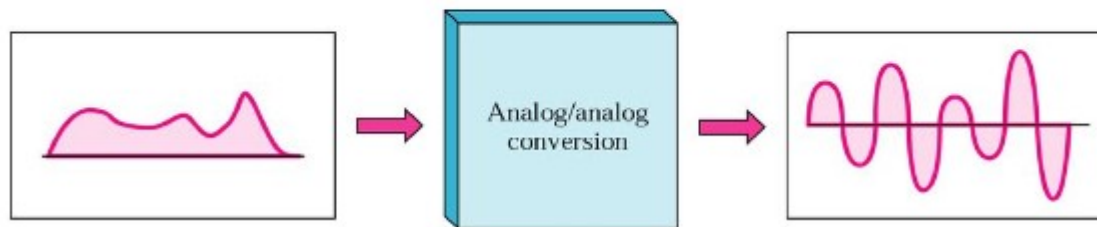
- If we have x variations in phase and y variation in amplitude, we will have $x*y$ possible variations. This is called Quadrature Amplitude Modulation (QAM).
- Possible variations of QAM are numerous.
- To avoid noise interference, QAM always uses more phase shifts than amplitude shifts.

Quadrature amplitude modulation is a combination of ASK and PSK so that a maximum contrast between each signal unit (bit, dibit, tribit, and so on) is achieved.

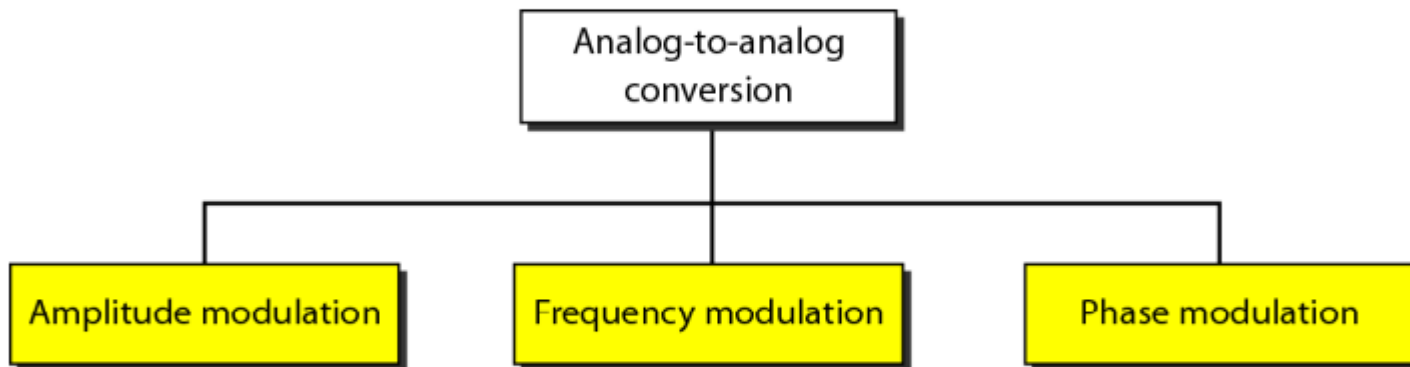
Analog-to- Analog conversion

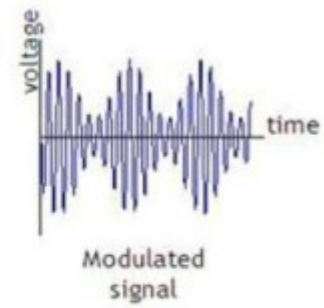
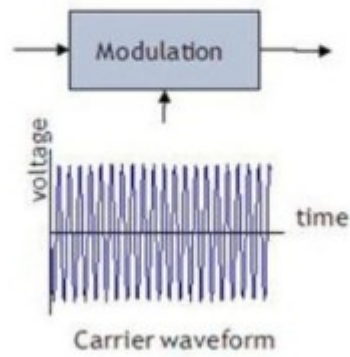
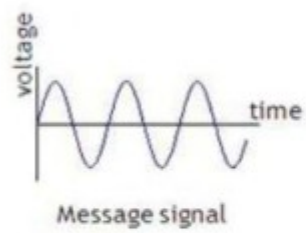
- Analog-to-analog conversion, or analog modulation, is the representation of analog information by an analog signal.
→modulate carrier frequency with analog data
- One may ask why we need to modulate an analog signal, it is already analog?
- Modulation is needed if the medium is bandpass in nature or if only a bandpass channel is available to us. modulate carrier frequency with analog data
- An example is radio. The government assigns a narrow bandwidth to each radio station.
- The analog signal produced by each station is a low-pass signal, all in the same range.
- To be able to listen to different stations, the low-pass signals need to be shifted, each to a different range.
- Analog-to-analog conversion can be accomplished in three ways: AM, FM and PM.

Analog-to-analog modulation



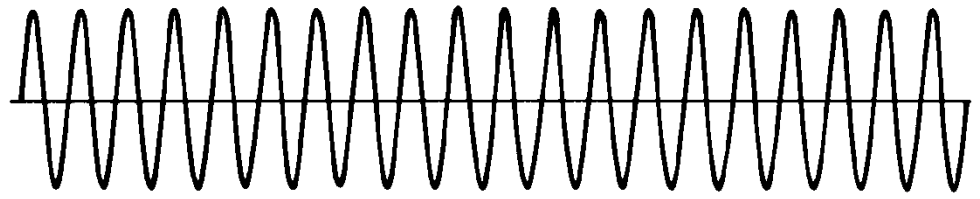
Types of analog-to-analog modulation



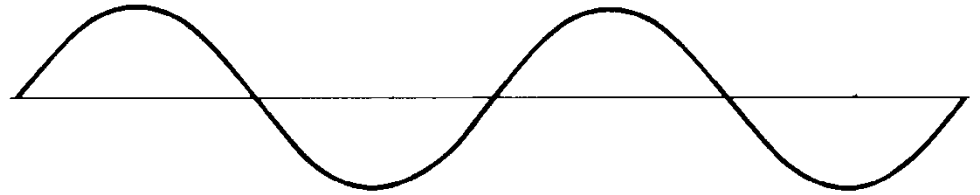


Analog Modulation Techniques

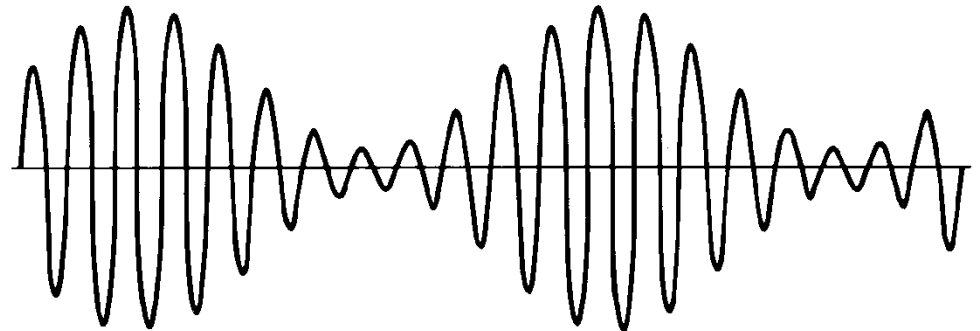
- Amplitude Modulation
- Frequency Modulation
- Phase Modulation



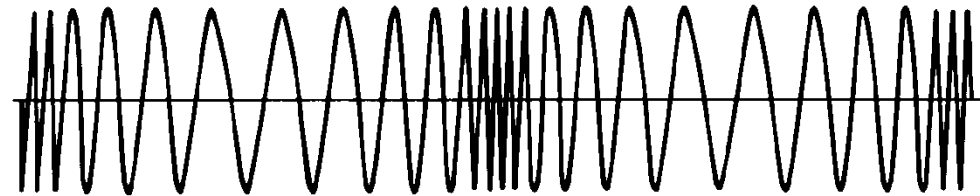
Carrier



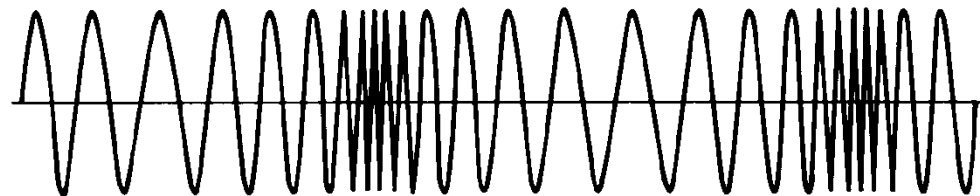
Modulating sine-wave signal



Amplitude-modulated (DSBTC) wave

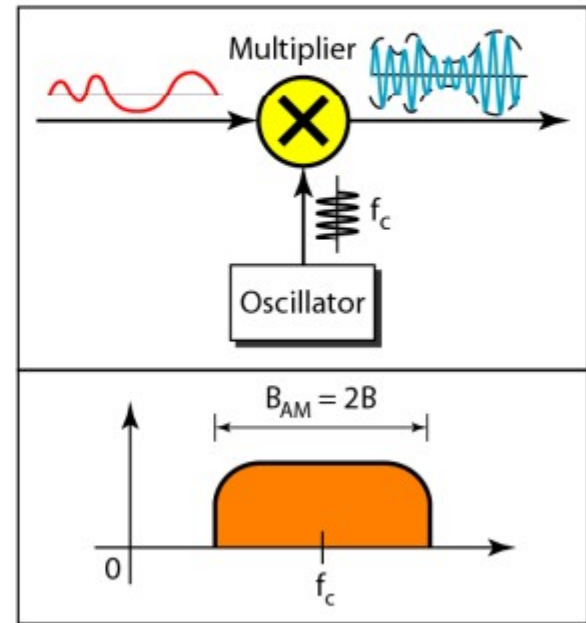
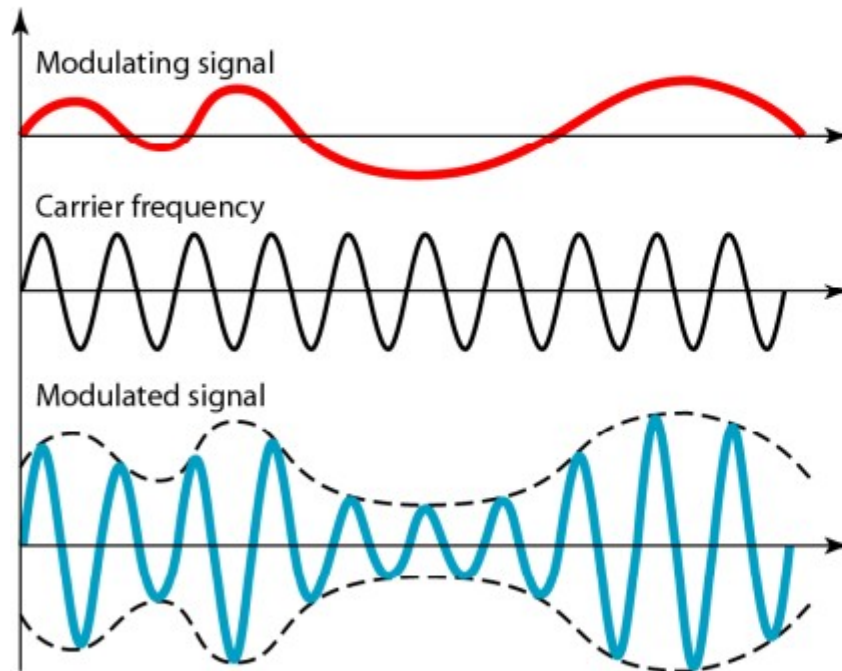


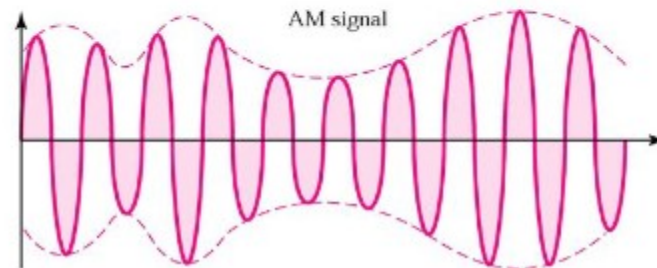
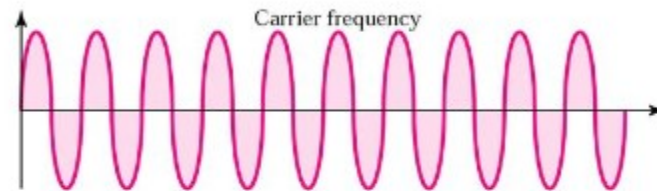
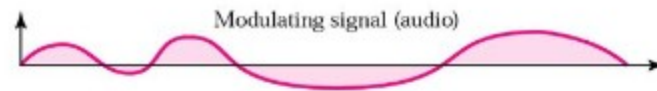
Phase-modulated wave



Frequency-modulated wave

Amplitude modulation

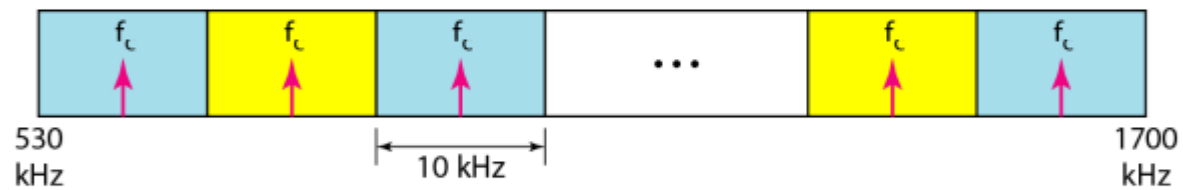




Note

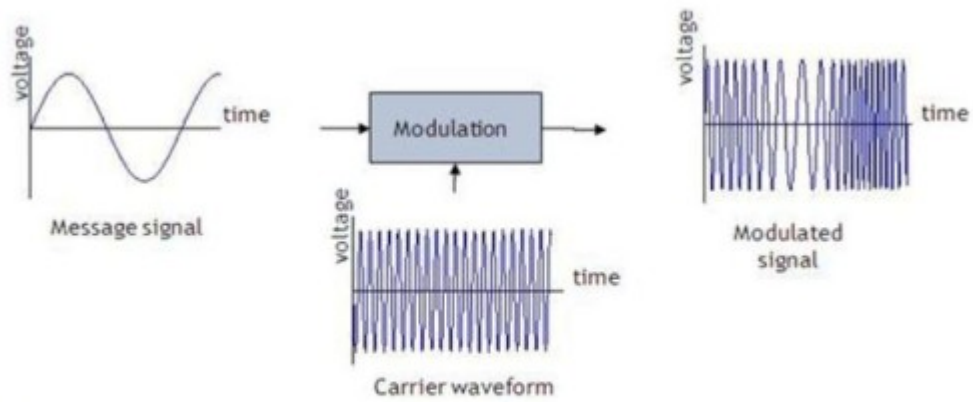
**The total bandwidth required for AM
can be determined
from the bandwidth of the audio
signal: $B_{AM} = 2B$.**

AM band allocation

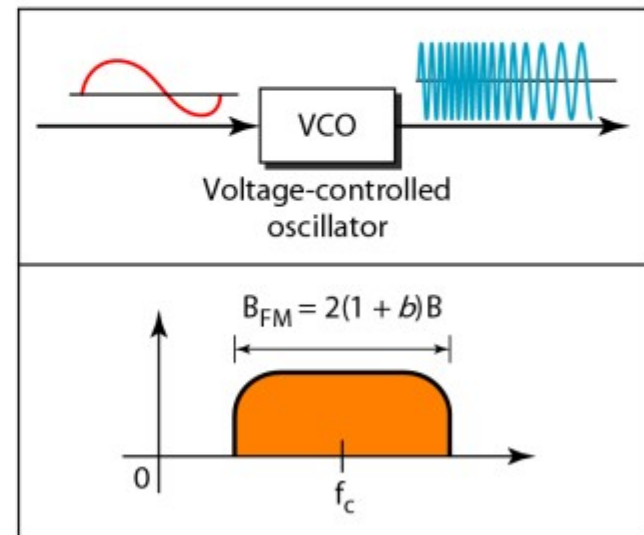
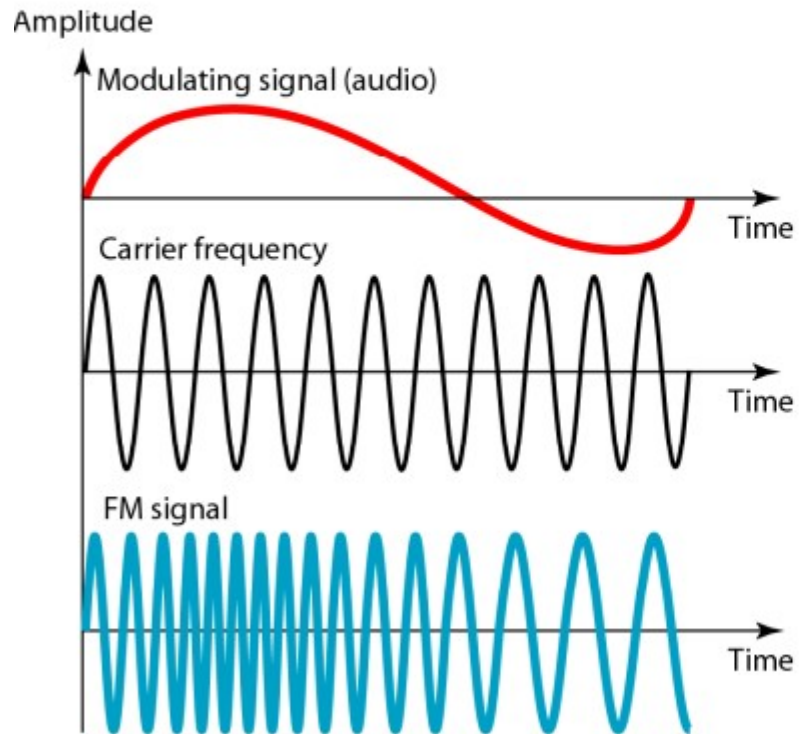


Frequency Modulation (FM)

- **Frequency Modulation (FM)**
In FM transmission, the frequency of the carrier signal is modulated to follow the changing voltage level (amplitude) of the modulating signal. The peak amplitude and phase of the carrier signal remains constant.
- The bandwidth of an FM signal is 10 times the bandwidth of the modulating signal.
- The bandwidth of an audio signal (speech and music) broadcast in stereo is almost 15KHz. Each FM station needs therefore a bandwidth of 150KHz.
- The FCC allows 200 KHz for each station to provide some room for guard bands.



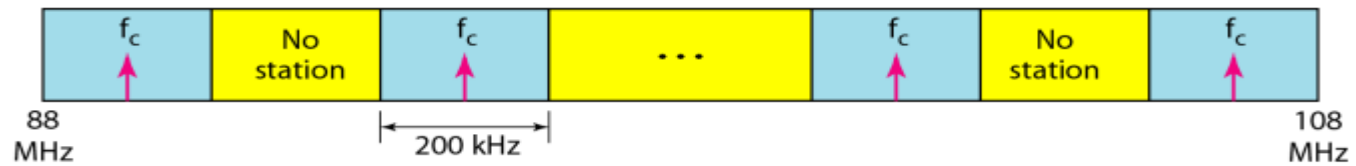
Frequency modulation



Note

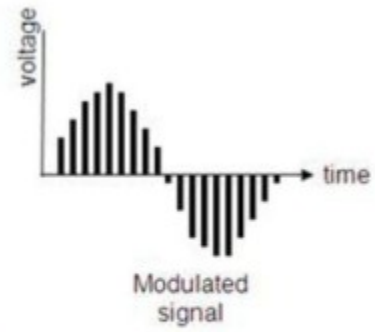
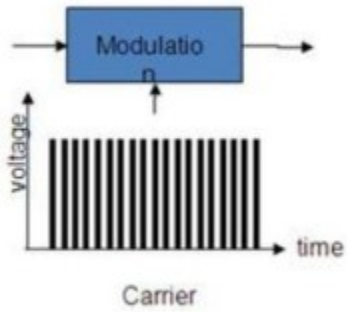
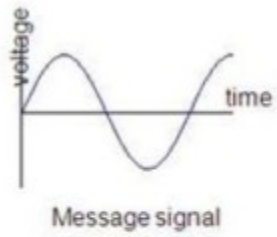
The total bandwidth required for FM can be determined from the bandwidth of the audio signal: $B_{FM} = 2(1 + \beta)B$.

FM band allocation

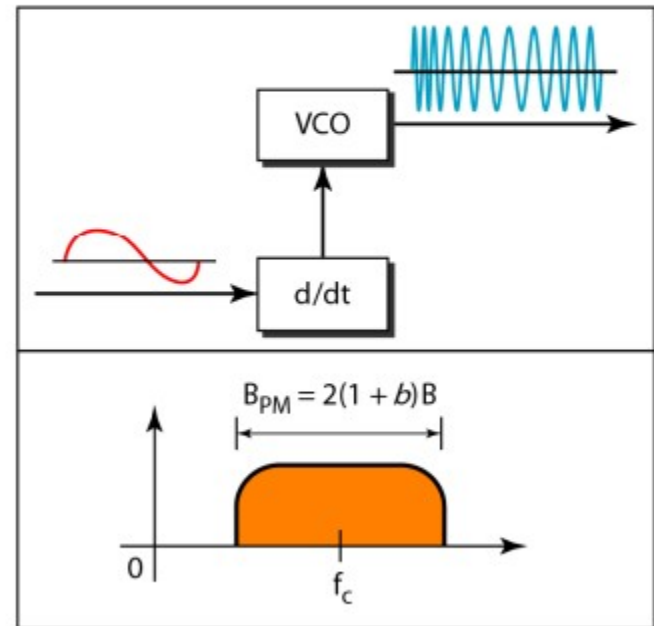
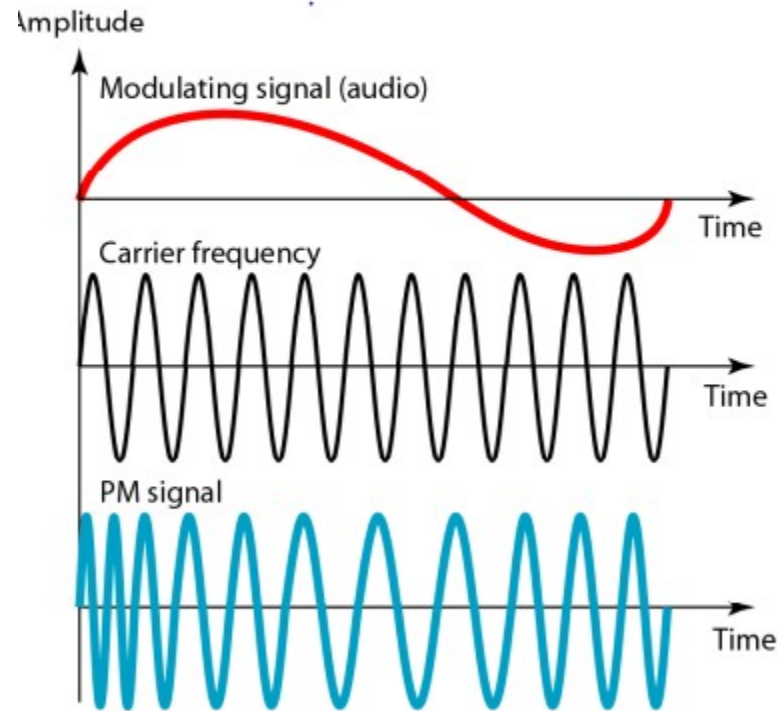


Phase Modulation (PM)

- Due to simpler hardware, PM is used in some systems as an alternative to FM.
- In PM transmission, the phase of the carrier signal is modulated to follow the changing voltage level (amplitude) of the modulating signal. The peak amplitude and frequency of the carrier signal remain constant.
- The analysis and final result (modulated signal) are similar to those of frequency modulation



Phase modulation



Note

The total bandwidth required for PM can be determined from the bandwidth and maximum amplitude of the modulating signal:

$$B_{PM} = 2(1 + \beta)B.$$